Monte Carlo Variance Reduction for Diffusion-Model Training: an Equal-Compute Comparison of Antithetic, Stratified and Importance Sampling

Research project report (single author)

Code, logs and every number: https://github.com/polylam1994/Monte-Carlo-variance-reduction-for-diffusion-model-trai

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Abstract

The DDPM training loss is a Monte Carlo average over data, timesteps and Gaussian noise, so standard variance-reduction methods from computational finance apply directly. We test whether they help, at equal compute, in a single controlled codebase. The arms are antithetic noise pairs $(\epsilon, -\epsilon)$, stratified and scrambled-Sobol timesteps, Improved-DDPM-style importance sampling of timesteps, and stratified timesteps with antithetic noise. Only the (x_0, t, ϵ) estimator changes, and every step uses 128 evaluations. We measure gradient variance directly at fixed parameters and decompose it by the law of total variance into between-timestep, data and noise parts. On CIFAR-10 the between-timestep part is only 6% of the total, so stratification removes little variance (geometric-mean ratio 0.82 [0.79, 0.86], with no measurable overhead). Antithetic pairs are strongly anti-correlated (pooled $\rho = -0.71$), but each pair shares one image and one timestep, and that cost cancels the gain (ratio 1.16 [0.98, 1.38] on CIFAR-10, 1.41 on Fashion-MNIST). The loss-based importance proposal over-weights low-noise timesteps. On CIFAR-10 it reduces variance early in training but increases it later (ratio 0.51–1.61 across checkpoints); a gradient-norm oracle proposal is predicted to give 0.62. On CIFAR-10 no estimator significantly changed held-out loss or FID after 3000 steps with 3 seeds (the detection limit is ≈1.7% / 12 FID). Importance sampling was best on both and nominally significant on Fashion-MNIST ($p = 0.04$, uncorrected), but not after correcting for multiple comparisons. Post-hoc probes show that a gradient-norm importance proposal would cut variance to ≈0.54× at fixed parameters. The decomposition predicts the sign of each effect before any training, which is the main practical result.

1 Introduction

Pricing an option by simulation and training a diffusion model have the same structure. In both, an expectation over random inputs is replaced by a sample average, and the error of that average limits progress. In finance, decades of work have produced methods that cut that error at almost no extra cost [3–5, 19]. Denoising score matching [8] is the same kind of average, over data, timesteps t and noise ϵ, and the gradient noise it injects into SGD is thought to slow training [11]. This report asks one concrete question: *does plugging these classical estimators into DDPM training reduce gradient variance, and does that make training faster or better at equal compute?*

What is and is not new. Importance sampling of timesteps is established [11, 16, 22, 23], and stratified or low-discrepancy timesteps are used in VDM [14] and CARV [2]. For these arms we replicate under controlled conditions. We did not find antithetic ϵ-pairs inside the DDPM training estimator in prior work: antithetic noise has been used at sampling time [1], and antithetic couplings for other gradient estimators [20]. The closest work on control variates for the DDPM gradient [9] found inconclusive effects on MNIST. Our contributions are: (1) an equal-compute comparison in one codebase where only the estimator changes, on Fashion-MNIST and CIFAR-10 with three seeds; (2) direct measurement of gradient variance at identical parameters for eight estimators, with a law-of-total-variance decomposition (Eq. 2) that *predicts* each estimator's variance; (3) a simple condition, $V_t + V_x < -\rho V_\epsilon$, for when antithetic noise can help, together with the measurement that

it fails here; (4) evidence that the loss-based importance proposal is mismatched to the gradient: the loss varies by ≈190× across timesteps, but gradient second moments by only ≈8×. A gradient-norm proposal measured post hoc reduces variance to ≈0.54×; (5) a finding that matters for anyone measuring gradient variance: single probe estimates are very noisy (coefficient of variation ≈20% at $R = 24$), so claims must rest on many probes. All numbers in this report are generated from logged runs by scripts in the repository.

2 Background

The DDPM objective as a Monte Carlo integral. With a fixed noise schedule $\bar{\alpha}_t$, $t \in \{0, \dots, T-1\}$, the ϵ-prediction loss of Ho et al. [8] is

$$\mathcal{L}(\theta) = \mathbb{E}_{x_0 \sim p_{\text{data}},\, t \sim \mathcal{U}\{0..T-1\},\, \epsilon \sim \mathcal{N}(0,I)} \big[\ell(\theta; x_0, t, \epsilon)\big], \qquad \ell = \tfrac{1}{d} \big\| \hat{\epsilon}_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon,\, t) - \epsilon \big\|^2. \tag{1}$$

Each optimiser step replaces $\nabla\mathcal{L}$ by an N-sample estimator $\hat{g} = \frac{1}{N} \sum_{i=1}^{N} w_i \nabla_\theta \ell(\theta; x_0^{(i)}, t^{(i)}, \epsilon^{(i)})$. The baseline draws all three inputs i.i.d. and sets $w_i = 1$. Pricing a derivative by simulation has the same form: an expectation of a payoff over random paths, estimated from a finite sample. Computational finance has a standard set of tools for this [4]: antithetic variates [3], stratification and quasi-Monte Carlo [19], importance sampling [5] and control variates [10]. All of them can be applied to $\hat{g}$ without changing $\mathbb{E}\hat{g} = \nabla\mathcal{L}$.

Where the variance comes from. Write $g = \nabla_\theta \ell(\theta; x_0, t, \epsilon)$ for a per-sample gradient and measure variance by the trace of the covariance, $\mathrm{V}(\cdot) = \mathbb{E}\| \cdot - \mathbb{E} \cdot \|^2$. The law of total variance splits it into three non-negative parts:

$$\mathrm{V}(g) = \underbrace{\mathrm{V}_t\big(\mathbb{E}[g \mid t]\big)}_{V_t\text{: between timesteps}} + \underbrace{\mathbb{E}_t\, \mathrm{V}_{x_0}\big(\mathbb{E}[g \mid x_0, t]\big)}_{V_x\text{: data, given } t} + \underbrace{\mathbb{E}_{x_0,t}\, \mathrm{V}_\epsilon\big(g \mid x_0, t\big)}_{V_\epsilon\text{: noise}}. \tag{2}$$

Each technique below acts on a different term. This is the main analytical tool of the paper: by measuring V_t, V_x and V_ϵ we can predict which technique can help before training.

3 Estimators

All estimators use exactly $N = 128$ evaluations of ℓ per step (an antithetic pair counts as two), are unbiased for $\nabla\mathcal{L}$, and differ only in how (x_0, t, ϵ, w) are drawn.

Stratified and quasi-Monte Carlo timesteps. `stratified` draws $u_i = (i + U_i)/N$, $U_i \sim \mathcal{U}[0,1)$, one point per stratum, and sets $t_i = \lfloor T u_i \rfloor$. Its variance is at most that of the baseline, because it removes the between-strata part of V_t: $\mathrm{V}(\hat{g}_{\text{strat}}) \approx (V_x + V_\epsilon + V_t^{\text{within}})/N$. `sobol` uses consecutive points of a scrambled one-dimensional Sobol sequence. For $N = 2^k$ each block of N points is a $(0, k, 1)$-net, so in one dimension it is essentially a stratified sample with a correlated scramble [17], and we expect `sobol`≈`stratified`. The "low-discrepancy sampler" of VDM [14] is a randomly shifted lattice from the same family.

Antithetic noise. `antithetic` draws $N/2$ triples (x_0, t, ϵ) and evaluates both ϵ and $-\epsilon$. With $\rho = \mathrm{Cov}(g(\epsilon), g(-\epsilon))/V_\epsilon$ (the trace correlation of the pair, given x_0, t),

$$\mathrm{V}(\hat{g}_{\text{anti}}) = \frac{2(V_t + V_x) + (1+\rho)V_\epsilon}{N} \quad \text{vs.} \quad \mathrm{V}(\hat{g}_{\text{unif}}) = \frac{V_t + V_x + V_\epsilon}{N}, \tag{3}$$

so antithetic sampling helps if and only if $V_t + V_x < -\rho\, V_\epsilon$. In option pricing the pair shares nothing but the normal draw, and the condition reduces to the textbook $\rho < 0$. Here the pair also shares (x_0, t), so it pays for the cancellation with half as many distinct images and timesteps. If $\hat{\epsilon}_\theta$ were affine in x_t, g would be affine in ϵ and $\rho = -1$: the odd part cancels exactly. The diagnostic estimator `paired_iid` uses the same pairing with two independent ϵ, so the gap between `paired_iid` and `antithetic` isolates the effect of ρ. `strat_anti` combines both ideas: the $N/2$ pair timesteps are stratified.

Importance-sampled timesteps. `importance` samples t from a piecewise-constant proposal q over 50 bins and reweights by $w = 1/(Tq(t))$, which leaves $\mathcal{L}$ unchanged [13]. Following Improved DDPM [16], $q(t) \propto \sqrt{\mathbb{E}[\ell^2 \mid t]}$, estimated from a 100-sample history per bin, used after every bin is filled, and mixed 90/10 with the uniform distribution so that $w \leq 10$. This proposal minimises the variance of the *loss* estimate. The proposal that minimises the trace of the *gradient* covariance is instead $q^\star(t) \propto \sqrt{\mathbb{E}[\|g\|^2 \mid t]}$ (the same Cauchy–Schwarz argument, applied to $\|g\|$), and it can differ substantially. We report the predicted variance under $q^\star$ as an oracle and do not train with it. `is_strat` pushes stratified uniforms through the inverse CDF of q (as in CARV [2]); it is probed but not trained.

4 Experimental setup

Model and training. DDPM with $T = 1000$, linear $\beta_t \in [10^{-4}, 0.02]$, ϵ-prediction and the unweighted loss (1). U-Net with residual blocks, GroupNorm, SiLU, sinusoidal time embedding and one self-attention layer at the lowest resolution. CIFAR-10 [15]: base width 32, multipliers $(1, 2, 2)$, two blocks per level, 1.54M parameters, random horizontal flips. (CIFAR-10 was loaded from the Hugging Face mirror `uoft-cs/cifar10`: 50,000/10,000 images, training-set pixel mean 120.708, which matches the standard channel means.) Fashion-MNIST [24]: width 32, multipliers $(1, 2)$, one block per level, 0.63M parameters. Adam [12], learning rate $2 \cdot 10^{-4}$ with 200 warm-up steps, gradient clipping at 1, batch $N = 128$, EMA 0.999 (with the standard warm-up on CIFAR-10), fp16 autocast on GPU. Everything except the estimator is identical across arms. For a given seed, every estimator starts from the same initialisation.

Budget and equal compute. We compare at an equal number of optimiser steps, which here also means an equal number of network evaluations: every step uses 128 forward/backward passes through the U-Net. CIFAR-10: 3,000 steps per run (384k evaluations, ≈7.7 epochs) on one NVIDIA T4. Fashion-MNIST: 800 steps per run on a 4-core CPU. Three seeds (0, 1, 2) per configuration, fixed in advance; no run was dropped. The whole study used 1.97 Kaggle GPU-hours of a 2-hour budget, including 0.42 GPU-hours lost to a failed data download (Appendix A). This budget, not the method, is the main limitation of the study (§6).

Metrics. (i) *Gradient variance at fixed parameters.* Inside each baseline run, at fixed checkpoints, we freeze θ and draw $R = 48$ independent mini-batch gradients from each of the eight estimators. x_0 is drawn from a fixed pool of 4,096 training images. We report $\operatorname{tr}\operatorname{Cov}(\hat{g})$ relative to `uniform` at the same θ. All estimators share the same checkpoints, so the ratio isolates the estimator. We also check unbiasedness: if two estimators have the same mean, $\mathbb{E}\|\bar{g}_e - \bar{g}_u\|^2 = (\operatorname{tr}_e + \operatorname{tr}_u)/R$; across all 98 estimator–checkpoint pairs the observed ratio of the two sides has mean 0.83 (range 0.34–2.50), so there is no sign of bias. (ii) *Decomposition* (2) in 20 timestep bins from paired mini-batches that share or resample (x_0, t) and ϵ, or flip the sign of ϵ (3 repetitions, $n = 64$ per batch). It gives V_t, V_x, V_ϵ and the pooled ρ, which we plug into (3) to *predict* each estimator's variance and compare with (i). (iii) *Held-out loss* on 2,000 test images with common random numbers: the same (x_0, t, ϵ) for every model and checkpoint, the standard variance-reduction device for comparing simulated alternatives [4]. (iv) *FID* [7] with clean-fid [18]: 10k samples, deterministic DDIM [21] with 25 steps from the EMA weights, and the same initial noise for every model, against CIFAR-10 training statistics. FID is computed at steps 1k/2k/3k for seed 0 and at 3k for seeds 1–2, for budget reasons. Absolute values are high because the models are small and briefly trained; only differences are interpreted. (v) *Overhead*: wall-clock per optimiser step, excluding evaluation. Differences are tested with two-sided Welch t-tests over seeds. With $n = 3$ these tests have low power, so we also report effect sizes.

5 Results

Gradient variance (Fig. 1, Table 1). Single probes are noisy (§5), so we summarise each estimator by the geometric mean of its ratio over all independent probes (seeds × checkpoints), with a 95% t-interval on the log scale and a sign test of "ratio < 1". *Stratification works, but the gain is small.* On CIFAR-10, stratified timesteps give 0.82 [0.79, 0.86], and every probe is below 1 (8/8, sign test $p = 0.008$). Scrambled Sobol timesteps are indistinguishable (0.83 [0.80, 0.86]), as expected in one dimension (H1). The gain is bounded

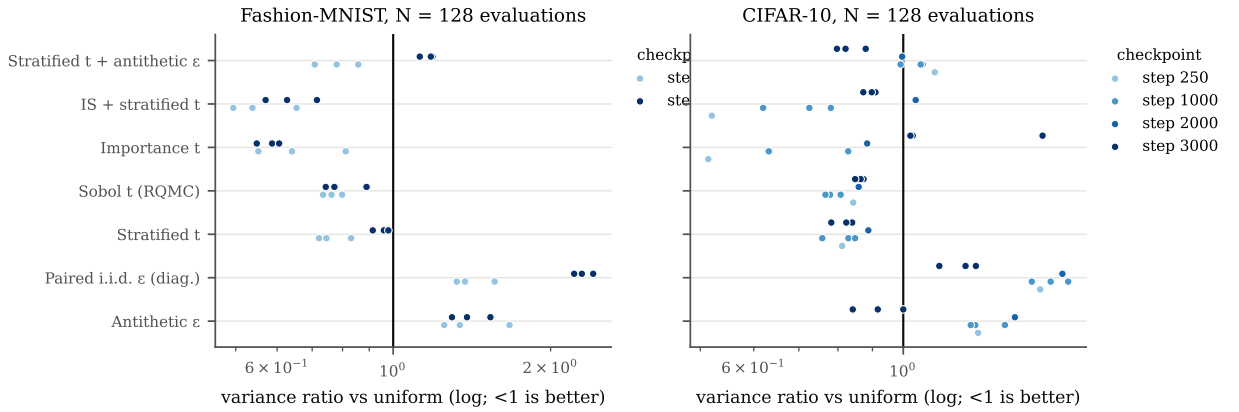


Figure 1: **Gradient variance relative to uniform sampling at the same parameters and the same $N = 128$.** Each dot is one (seed, checkpoint) probe with $R = 48$ independent mini-batch gradients per estimator; colour shades mark the checkpoint. Values below 1 mean less variance. Stratified and Sobol timesteps are below 1 in every plotted probe. Antithetic noise is mostly above 1, but below its paired-i.i.d. control in 8/8 (CIFAR-10) and 5/6 (Fashion-MNIST) probes. On CIFAR-10, importance sampling moves from well below 1 to above 1 as training proceeds.

Table 1: **Gradient-variance ratio vs. uniform** (tr Cov, same θ, $N = 128$), mean ± sd over seeds. CIFAR-10 steps 250/2000 are seed 0 only, steps 1000/3000 have 3 seeds; Fashion-MNIST has 3 seeds.

Estimator	fmnist @200	fmnist @800	cifar10 @250	cifar10 @1000	cifar10 @2000	cifar10 @3000	fmnist geo-mean [95% CI]	cifar10 geo-mean [95% CI]
Antithetic ϵ	1.42 ± 0.22	1.40 ± 0.12	1.29	1.32 ± 0.08	1.46	0.92 ± 0.08	1.41 [1.25, 1.58]	1.16 [0.98, 1.38]
Paired i.i.d. ϵ (diag.)	1.42 ± 0.13	2.31 ± 0.10	1.60	1.65 ± 0.10	1.72	1.22 ± 0.08	1.81 [1.35, 2.41]	1.47 [1.28, 1.70]
Stratified t	0.77 ± 0.06	0.95 ± 0.03	0.81	0.81 ± 0.05	0.89	0.82 ± 0.03	0.85 [0.74, 0.98]	0.82 [0.79, 0.86]
Sobol t (RQMC)	0.77 ± 0.03	0.80 ± 0.08	0.84	0.78 ± 0.02	0.86	0.86 ± 0.01	0.78 [0.73, 0.84]	0.83 [0.80, 0.86]
Importance t	0.67 ± 0.13	0.58 ± 0.03	0.51	0.70 ± 0.11	0.88	1.22 ± 0.34	0.62 [0.53, 0.72]	0.84 [0.62, 1.14]
IS + stratified t	0.56 ± 0.08	0.64 ± 0.07	0.52	0.71 ± 0.08	1.04	0.89 ± 0.02	0.60 [0.52, 0.69]	0.78 [0.64, 0.94]
Stratified t + antithetic ϵ	0.78 ± 0.07	1.16 ± 0.03	1.11	1.04 ± 0.04	1.00	0.83 ± 0.04	0.95 [0.75, 1.21]	0.96 [0.86, 1.07]

by the between-timestep share of the variance, only 6% (§5). *Antithetic noise does not reduce variance at equal compute.* On CIFAR-10 its ratio is 1.16 [0.98, 1.38] (above 1 in most probes, but not significantly: sign test $p = 0.29$). On Fashion-MNIST it is significantly worse, 1.41 [1.25, 1.58]. The coupling itself works: antithetic beats its paired-i.i.d. control, which differs only in the sign coupling, in 8/8 CIFAR-10 probes and 5/6 Fashion-MNIST probes (paired i.i.d.: 1.47 [1.28, 1.70]). What antithetic sampling cancels is roughly what it loses by sharing (x_0, t) within a pair. *Importance sampling depends on the training phase.* On CIFAR-10 the loss-based proposal ranges from 0.51 (first probe) to 1.61 (step 3000), with geometric mean 0.84 [0.62, 1.14]. Pushing stratified uniforms through the IS inverse CDF is the best timestep-only estimator we probed: 0.78 [0.64, 0.94] on CIFAR-10 and 0.60 [0.52, 0.69] on Fashion-MNIST. On Fashion-MNIST plain IS reduces variance at every probe (0.62 [0.53, 0.72]), so the late-training reversal is specific to CIFAR-10 in our data; the 800-step Fashion-MNIST runs may end before the proposal drifts. Stratified t with antithetic ϵ lands near 1 on both datasets (0.96 [0.86, 1.07]; 0.95 [0.75, 1.21]). The unbiasedness check passes for every estimator (§4).

Where the variance lives (Fig. 2, Table 3). On CIFAR-10 at step 3000, the per-sample gradient variance splits into 6% between timesteps (V_t), 44% from the data given t (V_x) and 50% from the noise (V_ϵ). This split explains the results above. Stratifying t can remove at most the 6% between-timestep part plus some within-bin variation. Antithetic noise attacks V_ϵ efficiently: $\rho = -0.71$, close to the -1 of a locally affine network. But by Eq. (3) it helps only if $V_t + V_x < -\rho V_\epsilon$, and here $V_t + V_x = 0.50$ while $-\rho V_\epsilon = 0.36$ (as fractions of the total). The condition fails, and it fails more clearly on Fashion-MNIST, where the data term dominates ($V_x = 63\%$, 0.69 vs 0.24). The decomposition gets the ranking of the estimators right and predicts the absolute uniform variance to within 10% on CIFAR-10 (less well on Fashion-MNIST, Table 3). It overestimates the antithetic penalty at step 3000 (predicted 1.15, measured 0.92 ± 0.08). For

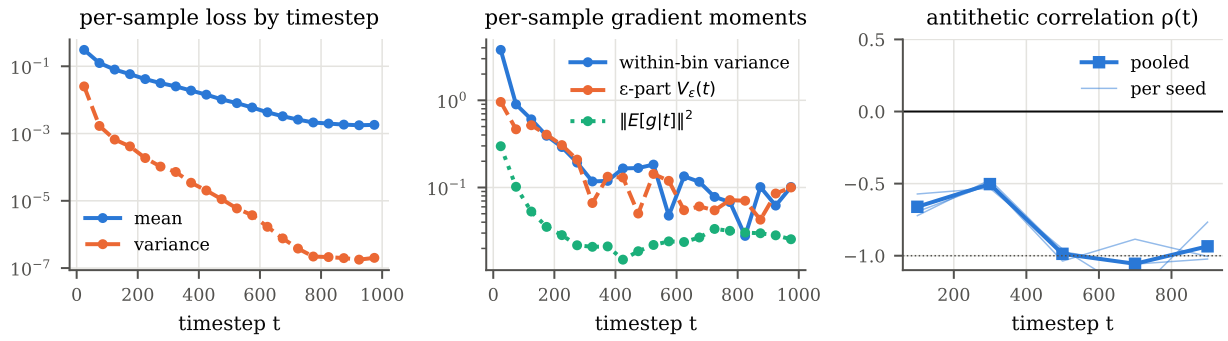


Figure 2: **Where the variance comes from (CIFAR-10, step 3000, 3 seeds).** Left: the per-sample loss and its variance fall by orders of magnitude from $t = 0$ to $t = T$. Middle: the per-sample gradient moments vary far less. Right: the correlation between the gradients for ϵ and $-\epsilon$ (pooled over 4 bins and over seeds; thin lines are single seeds) is strongly negative at all t.

Table 2: **Post-hoc re-probes at fixed parameters** (Fashion-MNIST, final checkpoints, CPU). Variance ratio vs. uniform: mean ± sd over 4 independent probe seeds ($R = 24$ each). Last column: prediction from the 8-repetition decomposition at the same checkpoint.

Estimator	seed s0, step 800	seed s1, step 800	seed s2, step 800	predicted (decomp.)
Antithetic ϵ	1.25 ± 0.22	1.14 ± 0.26	1.19 ± 0.24	1.41 / 1.48 / 1.38
Paired i.i.d. ϵ (diag.)	1.39 ± 0.28	1.29 ± 0.19	1.50 ± 0.14	1.61 / 1.67 / 1.59
Stratified t	0.82 ± 0.19	0.77 ± 0.16	0.76 ± 0.15	0.95 / 0.95 / 0.92
Sobol t (RQMC)	0.94 ± 0.26	0.91 ± 0.22	0.89 ± 0.13	–
Importance t	0.74 ± 0.11	0.71 ± 0.13	0.75 ± 0.13	–
IS + stratified t	0.70 ± 0.12	0.74 ± 0.19	0.72 ± 0.15	–
Stratified t + antithetic ϵ	1.02 ± 0.27	1.00 ± 0.23	0.95 ± 0.34	1.31 / 1.39 / 1.22
Oracle gradient-norm IS	0.53 ± 0.12	0.54 ± 0.07	0.54 ± 0.08	0.54 / 0.52 / 0.55

importance sampling, the loss-based proposal $q \propto \sqrt{\mathbb{E}[\ell^2 \mid t]}$ inherits the loss's ≈190× dynamic range across timesteps (ratio of the largest to smallest bin), whereas $\sqrt{\mathbb{E}\|g\|^2 \mid t}$ varies only ≈8×. The proposal therefore over-samples small t and gives large weights to the rest. The gradient-optimal proposal $q^\star$ would give a predicted ratio of 0.62 on CIFAR-10 (0.47 on Fashion-MNIST), the largest reduction available from any timestep-only method here. This is a prediction, not a trained result.

Probe noise and a direct test of the oracle proposal (post hoc, CPU). This analysis was added during self-review and was not pre-registered. On 3 saved Fashion-MNIST checkpoints (uniform runs, final step) we re-ran the probe 4 times with independent probe seeds ($R = 24$) and recomputed the decomposition with 8 repetitions instead of 3 (Table 2). The spread across repeats at *fixed* parameters is large: the median coefficient of variation of a single ratio is 20% (maximum 35%). Single-probe ratios, including the in-run ones, should therefore be read only in aggregate, as above. At the same checkpoints we also probed importance sampling with the gradient-norm proposal $q^\star \propto \sqrt{\mathbb{E}\|g\|^2 \mid t}$, estimated from the 8-repetition decomposition. Because $q^\star$ is estimated at the same parameters it is tested on, this is an oracle. Its measured ratio is 0.54 (range over checkpoints 0.53–0.54; predicted 0.54 / 0.52 / 0.55), against 0.73 for the loss-based proposal and 0.78 for stratification. The gradient-aware proposal is therefore a real, large variance reduction at fixed θ. Whether an online estimate of it improves *training* remains untested.

Table 3: **Law-of-total-variance decomposition and the variance ratios it predicts** (Eq. 3), compared with the direct measurement at the same checkpoint. The last column is the absolute per-sample variance of the uniform estimator ($N \cdot \mathrm{tr}\,\mathrm{Cov}$), predicted and measured.

	stratified	antithetic	paired i.i.d.	strat+anti	oracle grad-IS	$N\,\mathrm{tr}\,\mathrm{Cov}$ (unif.)
Fashion-MNIST, step 800, 3 seed(s)						
fractions $V_t/V_x/V_\epsilon$	6% / 63% / 31%	pooled ρ = -0.78 ± 0.12		$V_t + V_x$ = 0.69 vs $-\rho V_\epsilon$ = 0.24		
predicted ratio	0.94 ± 0.02	1.45 ± 0.08	1.69 ± 0.04	1.33 ± 0.10	0.47 ± 0.04	2.115 ± 0.214
measured ratio	0.95 ± 0.03	1.40 ± 0.12	2.31 ± 0.10	1.16 ± 0.03	–	1.655 ± 0.083
CIFAR-10, step 3000, 3 seed(s)						
fractions $V_t/V_x/V_\epsilon$	6% / 44% / 50%	pooled ρ = -0.71 ± 0.05		$V_t + V_x$ = 0.50 vs $-\rho V_\epsilon$ = 0.36		
predicted ratio	0.94 ± 0.02	1.15 ± 0.05	1.50 ± 0.03	1.03 ± 0.05	0.62 ± 0.03	0.405 ± 0.011
measured ratio	0.82 ± 0.03	0.92 ± 0.08	1.22 ± 0.08	0.83 ± 0.04	–	0.438 ± 0.012

Table 4: **Training outcomes on CIFAR-10 at equal steps (3000) and equal evaluations per step.** Mean ± sd over seeds. p: two-sided Welch t-test against uniform (pre-registered). FID@1000/2000 are seed 0 only. The `strat_anti` seed-2 run was trained without FID (PLAN A5), so its FID uses 2 seeds. ms/step excludes evaluation; T4 GPU, fp16.

Estimator	seeds	ms/step	overhead	held-out ($\times 10^{-3}$)	p	FID@1000	FID@2000	FID@3000	p (FID)
Uniform (baseline)	3	90.8	+0.0%	37.50 ± 0.18	–	271.7	191.1	141.0 ± 5.7	–
Antithetic ϵ	3	91.0	+0.2%	37.75 ± 0.02	0.14	269.6	190.7	143.3 ± 3.0	0.58
Stratified t	3	90.4	-0.5%	37.70 ± 0.22	0.29	270.2	190.5	140.0 ± 2.8	0.80
Importance t	3	92.2	+1.5%	36.92 ± 0.32	0.07	269.1	184.8	135.3 ± 4.1	0.24
Stratified t + antithetic ϵ	3	89.4	-1.6%	37.65 ± 0.19	0.37	271.7	189.3	141.2 ± 3.7	0.96

Training outcomes (Table 4, Figs. 3–4). *No estimator changes training outcomes significantly* (with 3 seeds, differences smaller than ≈1.7% in held-out loss or ≈12 FID are below the detection limit; §6). At step 3000 the held-out loss ($\times 10^{-3}$) is 37.50 ± 0.18 for uniform, 37.70 ± 0.22 for stratified ($p = 0.29$), 37.75 ± 0.02 for antithetic ($p = 0.14$), 37.65 ± 0.19 for stratified+antithetic ($p = 0.37$) and 36.92 ± 0.32 for importance sampling ($p = 0.07$). FID-10k is 141.0 ± 5.7 (uniform), 140.0 ± 2.8, 143.3 ± 3.0, 141.2 ± 3.7 and 135.3 ± 4.1 respectively (all $p \geq 0.24$). The 15–20% variance reduction from stratification does not register in either metric. Its effect on the held-out loss (+0.5%) is far below the seed spread. Importance sampling is best on both metrics despite *higher* gradient variance late in training. In a secondary analysis that pairs runs by seed (same initialisation and the same evaluation noise; not pre-registered), the differences are -1.5 ± 0.8% in held-out loss ($p = 0.08$) and -5.7 ± 2.7FID ($p = 0.06$). This is suggestive but not significant. Gradient clipping was never active after warm-up in any arm (logged gradient norms $\ll 1$), so clipping bias does not explain it. Two *conjectures* we could not test within budget: (a) the large early-training variance reduction (0.51) gives IS a head start that persists; (b) because Adam normalises each coordinate, re-allocating samples towards small t changes the effective step size per timestep, even though the gradient estimator is unbiased. *Overhead is negligible.* All estimators run at 90.8–92.2 ms/step (within ±3% of uniform, inside timing noise), so loss vs wall-clock and loss vs steps are the same comparison.

Fashion-MNIST. The CPU runs (800 steps, 3 seeds; Appendix C) agree with CIFAR-10 for stratification and antithetic noise. Held-out loss ($\times 10^{-3}$) is 56.93 ± 0.66 (uniform), 56.66 ± 0.77 (stratified, $p = 0.67$), 57.89 ± 0.87 (antithetic, $p = 0.21$) and 57.10 ± 0.54 (strat+anti, $p = 0.75$). Importance sampling, which reduces gradient variance at every Fashion-MNIST probe, gives the one nominally significant result of the study: 55.41 ± 0.57, $p = 0.04$ (Welch); paired by seed -2.7 ± 0.2%, $p = 0.002$. Two caveats. It does not survive a Bonferroni correction over the four comparisons per dataset ($\alpha = 0.0125$). And our pre-registered criterion for "helps training" also required an FID improvement, which is not available for Fashion-MNIST. Taken together with the same direction on CIFAR-10 ($p = 0.07$), it is consistent, but not conclusive, evidence that loss-based importance sampling improves the held-out loss by 1.5–3% at equal steps in short training runs.

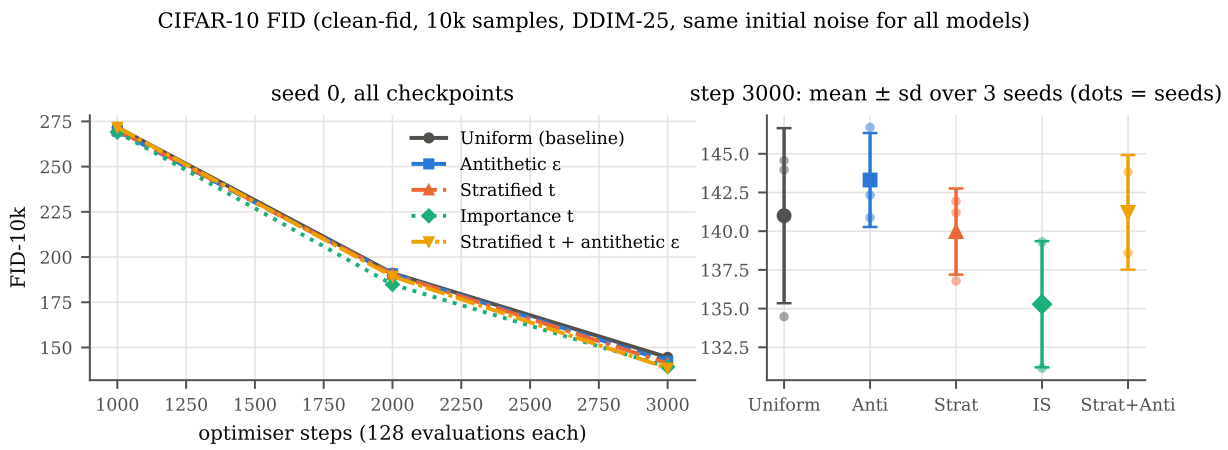


Figure 3: **FID vs training steps (CIFAR-10).** Left: all checkpoints for seed 0. Right: step 3000, mean ± sd over seeds (dots are individual seeds). All estimators lie within one seed-to-seed standard deviation of uniform.

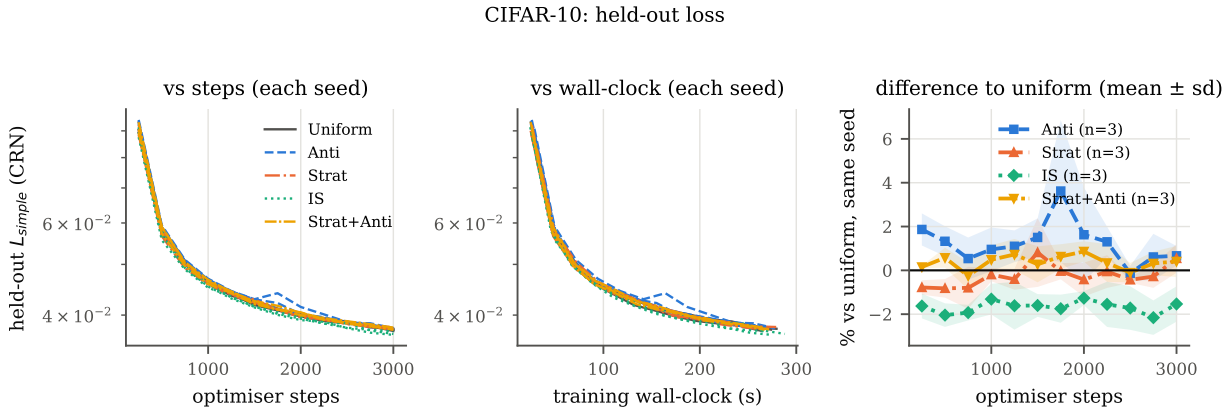


Figure 4: **Held-out loss (common random numbers) vs steps and vs wall-clock, CIFAR-10.** Right: relative difference to uniform for the same seed (same initialisation), mean ± sd. Importance sampling is consistently about 1.5% lower. The difference is not significant under the pre-registered test.

6 Limitations

- **Scale.** The models are small (1.5M / 0.6M parameters) and trained briefly (3000 steps on CIFAR-10, 800 on Fashion-MNIST) because of the 2-GPU-hour budget. FIDs around 140 mean the models are far from converged. Gradient-noise effects may matter more, or less, late in training and at scale; we cannot tell which. The variance *measurements*, in contrast, are exact statements about these estimators at these parameters.
- **Power.** With three seeds per arm, a two-sided test at $\alpha = 0.05$ has 80% power only for effects of about 3.1 within-arm standard deviations. Here that is ≈1.7% in held-out loss or ≈12 FID points. A non-significant result therefore rules out only large effects; it is not evidence of equivalence. The importance-sampling difference (-1.5% held-out loss) is just below this threshold.
- **FID protocol.** Because of the budget, FID was computed at three checkpoints for seed 0 and only at the final checkpoint for seeds 1–2 (PLAN amendment A4). We used DDIM-25 rather than DDIM-50, 10k rather than 50k samples, and training-set statistics. All of these are fixed across arms, but they

make the absolute values incomparable to the literature.

- **One optimiser, one weighting.** Everything uses Adam with L_{simple}. Adam's per-coordinate normalisation may absorb part of the benefit of lower gradient variance [9]. Other loss weightings (e.g. ELBO or Min-SNR [6]) change the per-timestep profile, and with it V_t, and could favour stratification or importance sampling more.
- **Importance proposal.** We trained only the Improved-DDPM loss-based proposal. The gradient-norm oracle is evaluated analytically from the decomposition, not trained. Its predicted gain is a conjecture until it is trained with an online estimate of $\mathbb{E}[\|g\|^2 \mid t]$.
- **Decomposition noise.** Per-bin estimates use 3 repetitions of single mini-batch differences, so individual bins are noisy (single-bin ρ can fall outside $[-1, 1]$). We report ratio-of-sums pooled estimates. The predicted uniform-estimator variance matches the direct measurement to within a few percent on CIFAR-10 but less well on Fashion-MNIST (Table 3).
- **Control variates** (item 5 of the original brief) were not implemented.

7 Conclusion

Classical variance reduction transfers to diffusion training in the sense that the estimators are unbiased, cheap and act as theory predicts. In this regime they do not translate into better models at equal compute. The law-of-total-variance decomposition explains why. The DDPM gradient's variance is dominated by data and noise *within* a timestep, not by the choice of timestep, which caps what stratification can achieve. Antithetic noise cancels the noise term efficiently, but pays for it by halving the number of distinct images. The finance lesson that transfers is diagnostic: measure the variance components first, then choose the estimator. The most promising direction the measurements point to is a gradient-norm-based importance proposal, measured post hoc at $\approx 0.54\times$ the baseline variance at fixed parameters, combined with stratification. The only training-outcome signal, a 1.5–3% held-out-loss gain from loss-based importance sampling, is consistent across both datasets but not conclusive. That is a conjecture to test at larger scale and with more seeds.

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A Compute budget and provenance

All GPU work ran as private Kaggle script kernels on T4×2 machines, with two independent runs in parallel, one per GPU. GPU usage was read from the Kaggle quota API before and after every session (`results/budget.json`). Total: 1.97 GPU-hours of the 2.0 available. The first smoke test spent its entire 25-minute session downloading CIFAR-10 from the authors' server at ∼90 kB/s (0.42 GPU-hours, no training). We then switched to the Hugging Face mirror of the same images. Each kernel embeds the exact source files and prints their SHA-256 hashes, which match the git commit named in every run's `meta.json`. Results return through the kernel log as a checksummed archive (`scripts/fetch_kaggle_results.py`). Every run is listed in `results/runs.csv` with its configuration, seed, runtime, metrics and kernel URL. Every table, figure and in-text number is regenerated by `scripts/make_tables.py`, `make_figures.py` and `make_numbers.py`.

B Pre-registered plan and deviations

`PLAN.md` was committed before any run. All deviations are listed there as amendments A1–A5, with their reasons: model width 64→32 and DDIM-50→25 after measured timings, EMA warm-up, Fashion-MNIST schedule, a reduced FID schedule for seeds 1–2 (A4), and a training-only run for `strat_anti` seed 2 (A5). No seed or run was excluded. The post-hoc CPU analyses in §5 were added during self-review and are labelled as exploratory.

C Fashion-MNIST results

Table 5: Fashion-MNIST, 800 steps, CPU. Mean ± sd over seeds; p: Welch t-test vs uniform. ms/step is for a single-threaded worker (4 workers ran in parallel).

Estimator	seeds	ms/step	overhead	held-out ($\times 10^{-3}$)	p
Uniform (baseline)	3	3111.9	+0.0%	56.93 ± 0.66	–
Antithetic ϵ	3	3123.1	+0.4%	57.89 ± 0.87	0.21
Stratified t	3	3090.0	-0.7%	56.66 ± 0.77	0.67
Importance t	3	3096.0	-0.5%	55.41 ± 0.57	0.04
Stratified t + antithetic ϵ	3	3050.7	-2.0%	57.10 ± 0.54	0.75

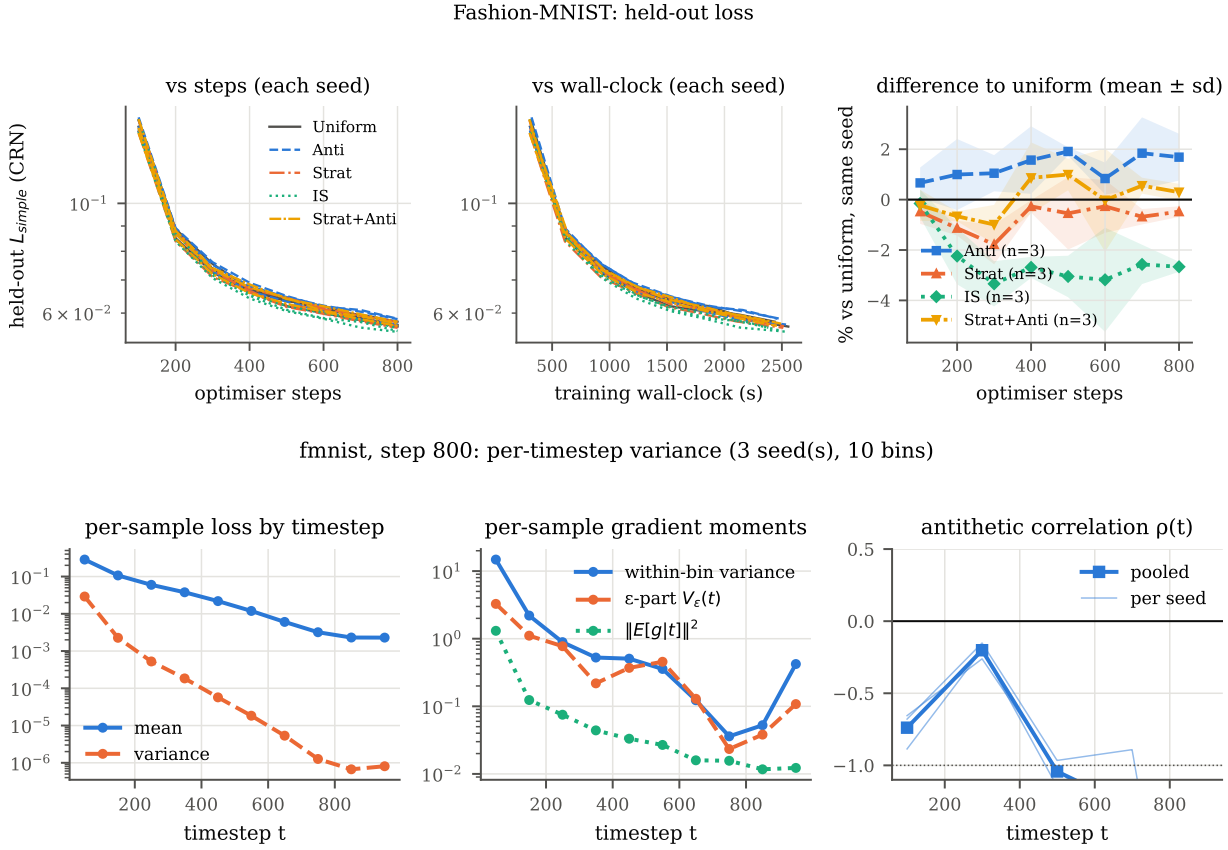


Figure 5: Fashion-MNIST held-out loss and per-timestep decomposition.